Advanced SQL Assignment Questions

Q1. What is a Common Table Expression (CTE), and how does it improve SQL query readability?

Common table of expression (CTE) it is temporary result set define within SQL statement.

- Improve readability and modularity
- Allows reuse of intermediate results.
- Makes recursive queries easier to write.

Q2. Why are some views updatable while others are read-only? Explain with an example.

Update views = based on a single table without aggregation, join. Change the view reflect in the base table.

Read-only view = contain join, aggregate or grouping making update ambiguous.

Q3. What advantages do stored procedures offer compared to writing raw SQL queries repeatedly?

- Improved performance
- Reusable & maintainable logic
- Reduce network load
- Enhance security
- Reduce repetition code in SQL

Q4. What is the purpose of triggers in a database? Mention one use case where a trigger is essential.

- Trigger automatically execute in response to certain events (insert, update, delete).
- Audit changes or enforce business rules automatically.

Q5. Explain the need for data modelling and normalization when designing a database.

Data modelling is the process designing the structure and organization the data within database.

Defines what data is stored, how different pieces of data relate, and the rules governing them.

Normalization reduce redundancy and improved consistency.